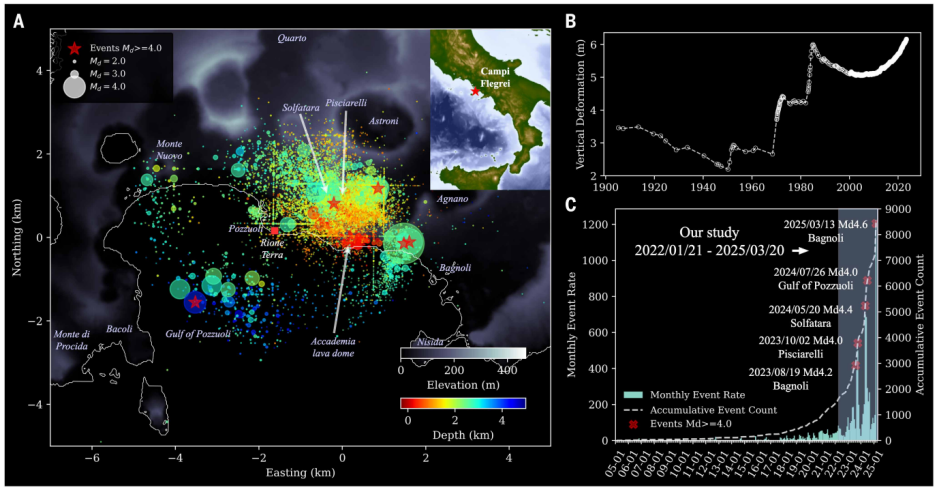
Figure 1: Seismicity and ground displacement

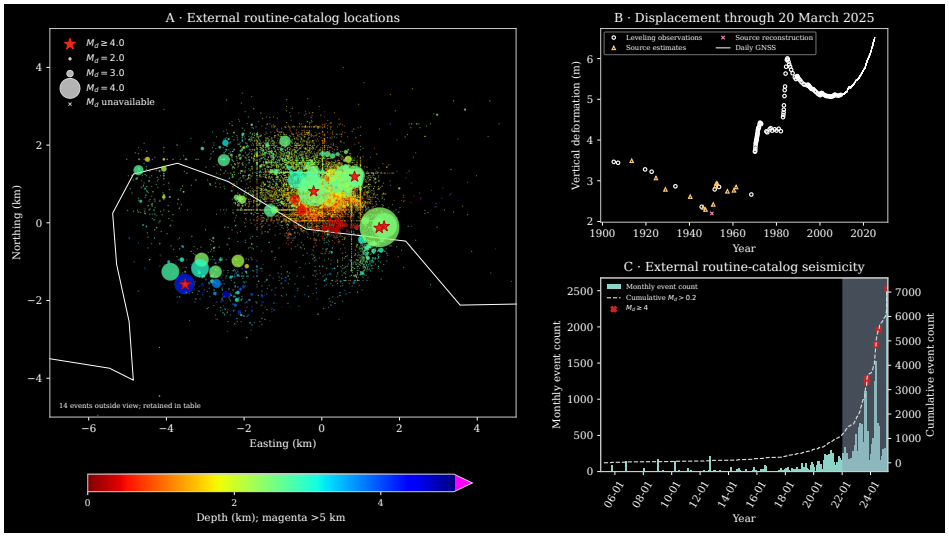
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

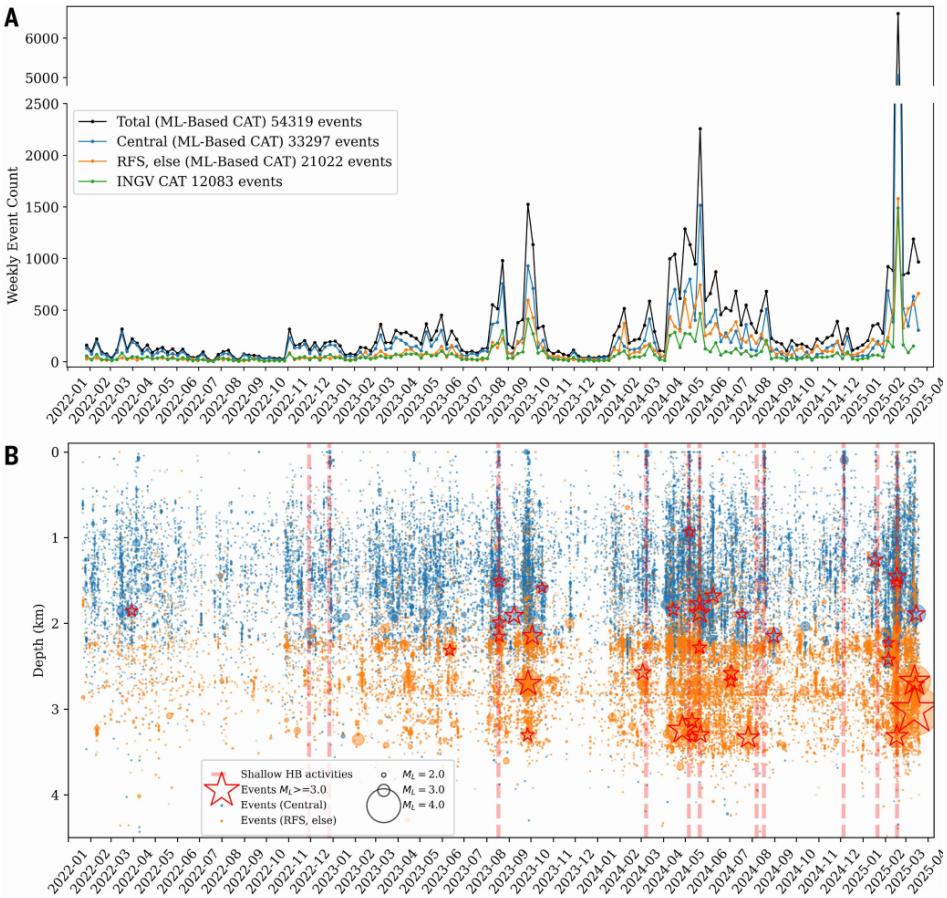
Panel 1A–C — partially reproduced



External routine-catalog context and source-labeled leveling/GNSS observations. These earthquakes are not independent detections. Historical snapshot and datum differences remain.

Figure 2: Temporal evolution of seismicity

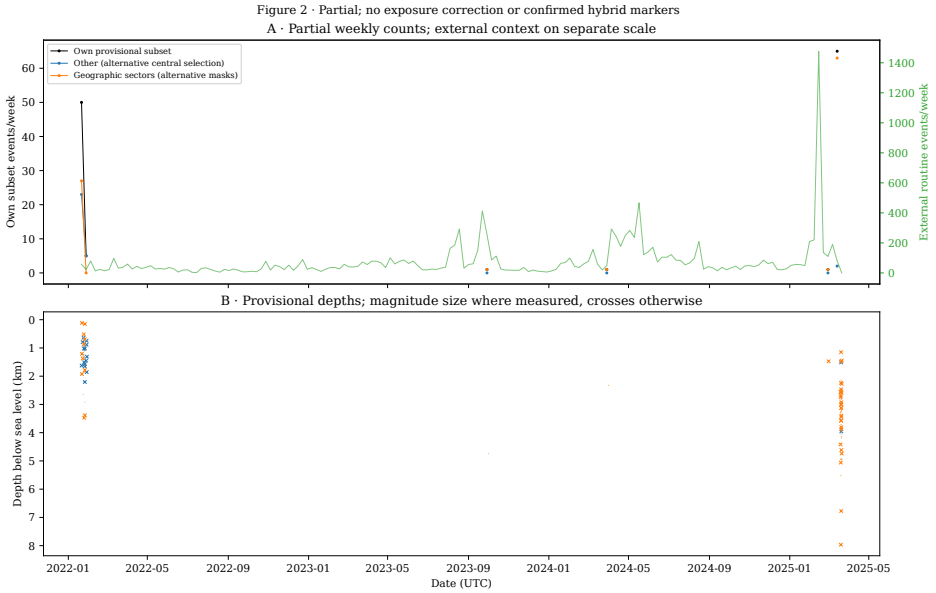
Published reference



Tan et al. (2025), tanmodel.pdf, p. 3.
DOI: 10.1126/science.adw9038.

Independent result

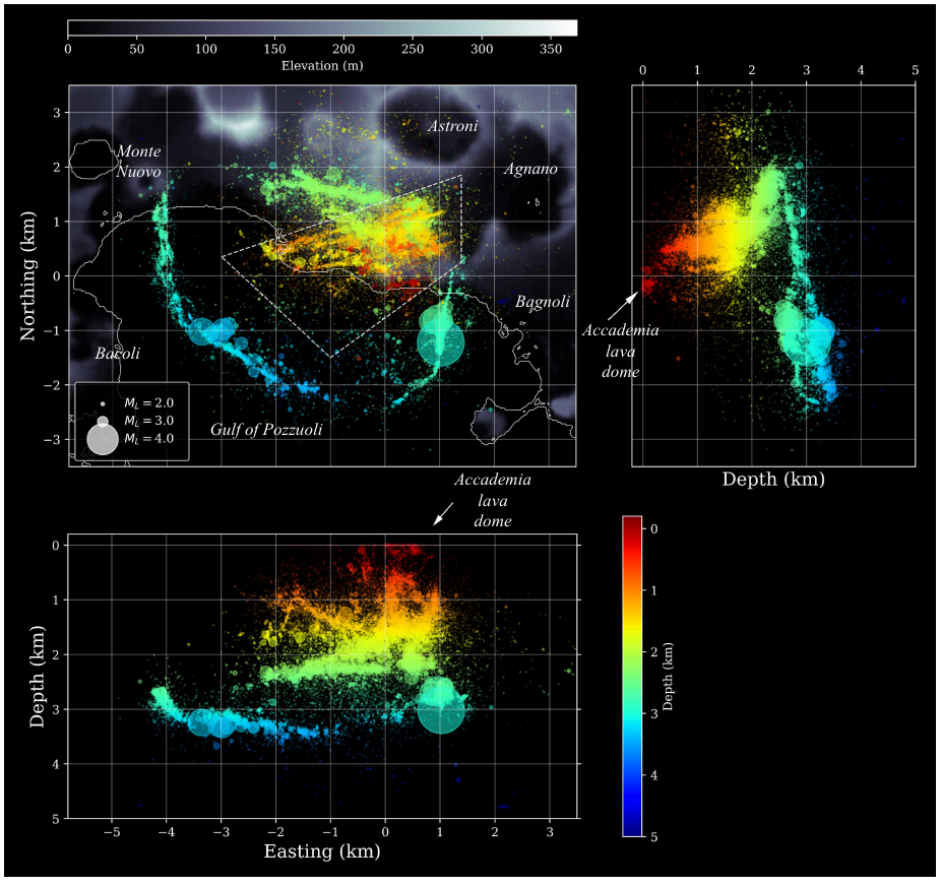
Panel 2; latest provisional numerical counterpart — partially reproduced



Weekly counts and depths from the latest 123-event relocated trial. Weeks without observations are not connected as measured zero rates. Routine counts are separately labeled context; geographic masks are explicit alternatives.

Figure 3: Spatial distribution of seismicity

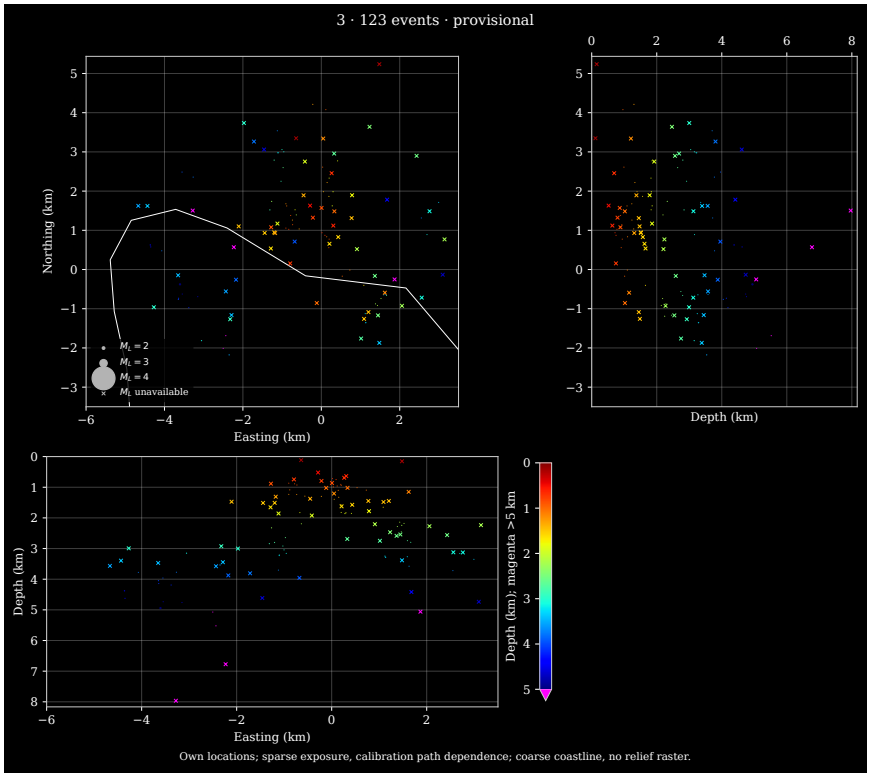
Published reference



Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

Independent result

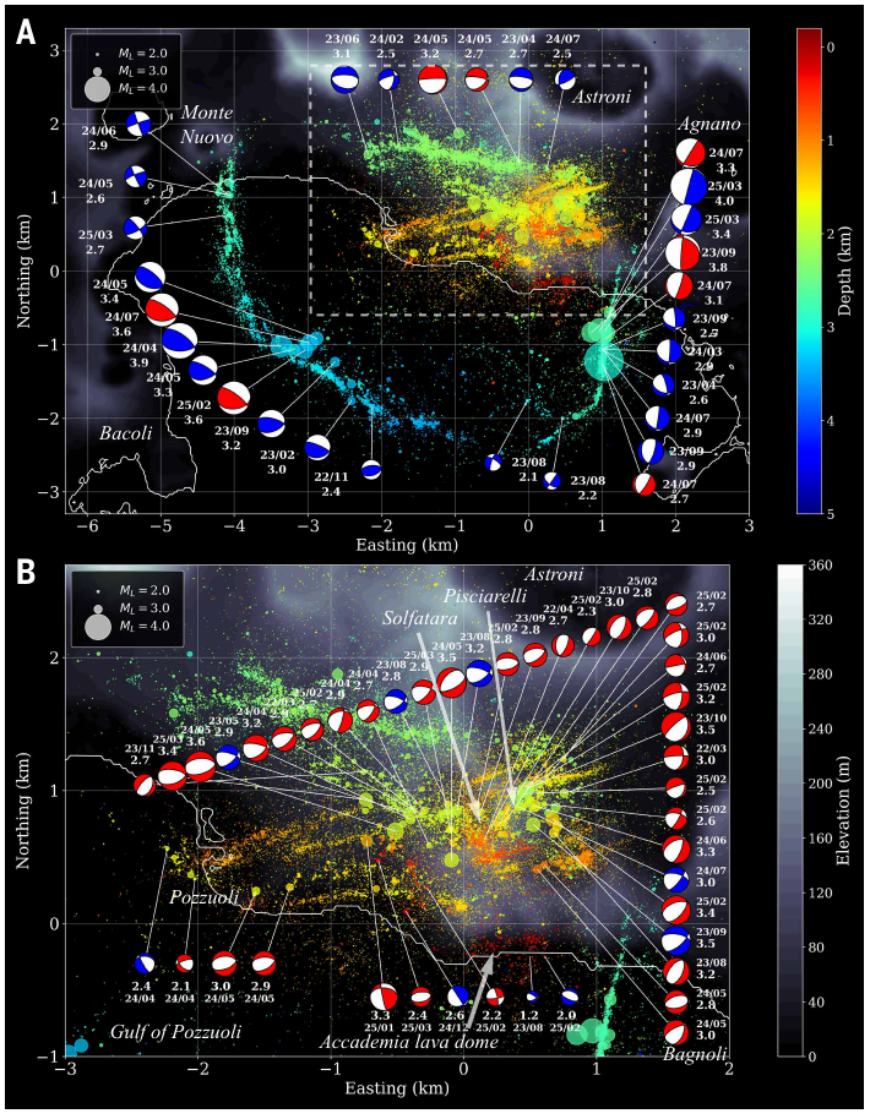
Panel 3; latest provisional numerical counterpart — partially reproduced



123 independently relocated trial events, 0.115–7.965 km below sea level. Own provisional magnitudes set point areas where available; crosses indicate missing magnitudes. Five depths above 5 km retain magenta over-range colors. This subset does not establish a ring fault or its depth limits.

Figure 4: Focal mechanisms of larger events

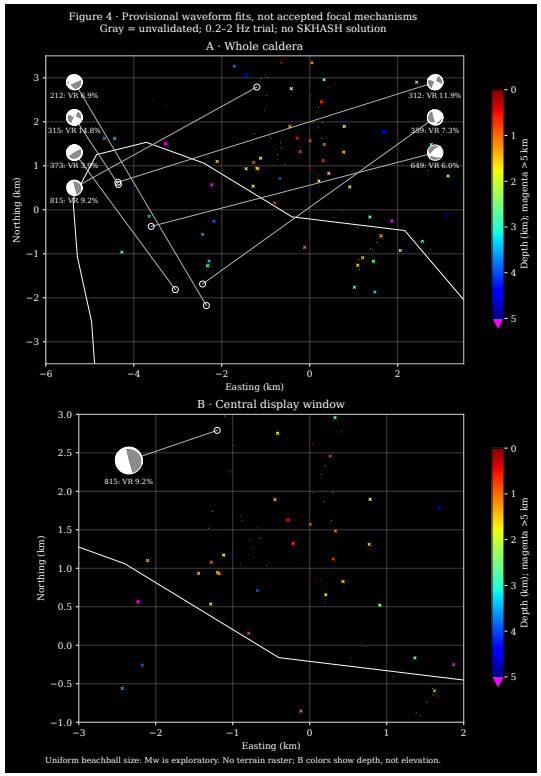
Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

Panel 4A–B; unvalidated waveform-inversion trials — partially reproduced



Seven independently computed but unaccepted trial fits. Poor waveform fit and 34–98 degree frequency-band sensitivity prevent mechanism interpretation. Gray denotes unvalidated; zero SKHASH solutions. B uses depth rather than unavailable elevation. Frozen candidate cohort.

Figure S1: Recorded station availability

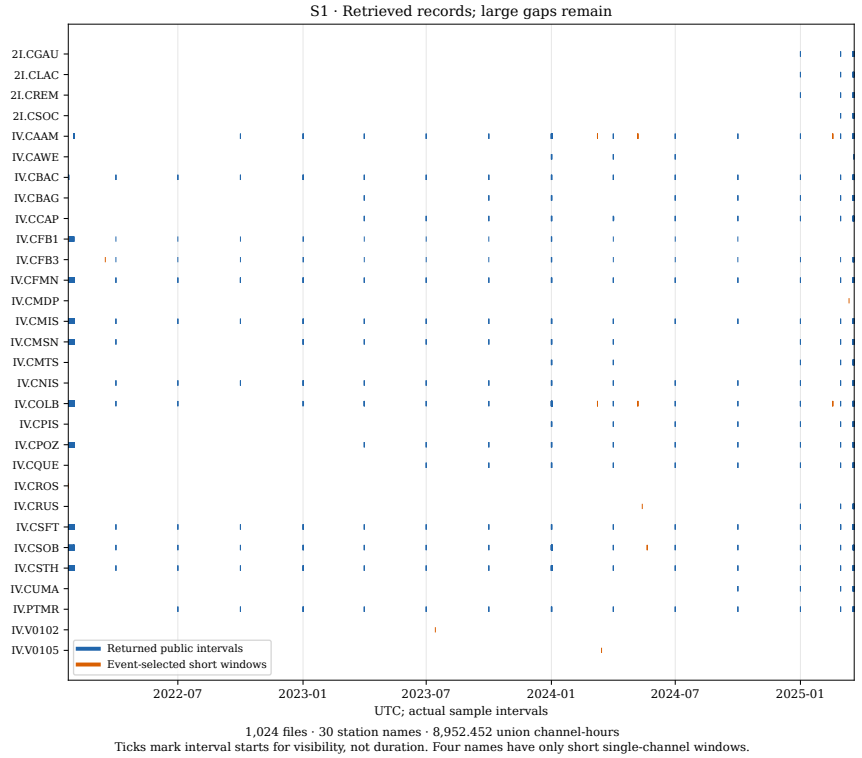
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

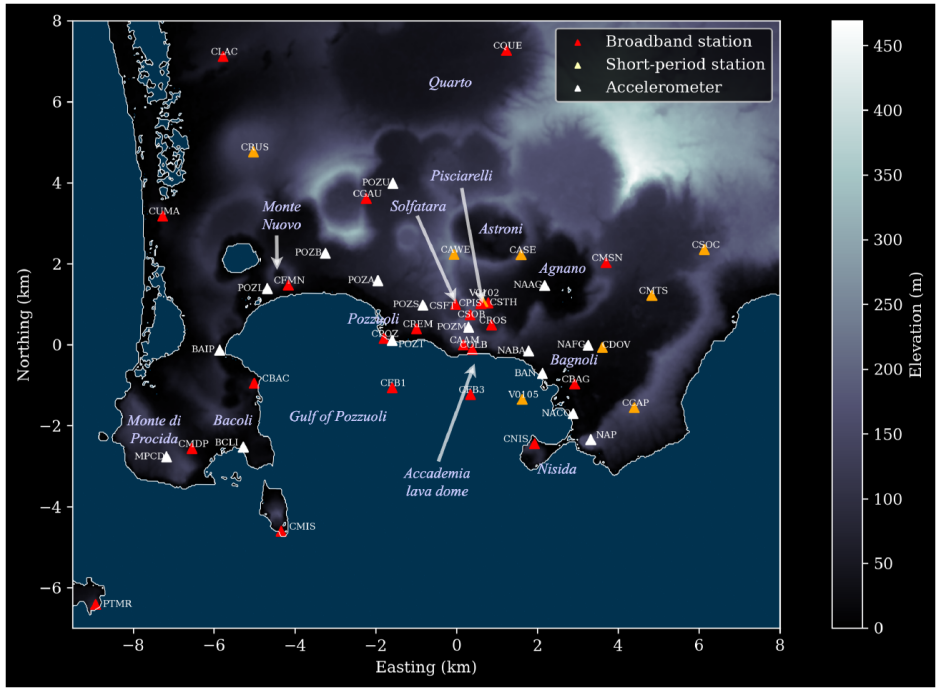
Panel S1; actual returned intervals — partially reproduced



1,024 waveform files, 30 station names, 8,952.452 union channel-hours. Four station names occur only in short single-channel windows. Orange identifies event-selected observations; neither these windows nor requested-but-unreturned intervals establish continuous coverage. Official response epochs cover 8,952.386 hours; four short intervals lack responses.

Figure S2: Station locations

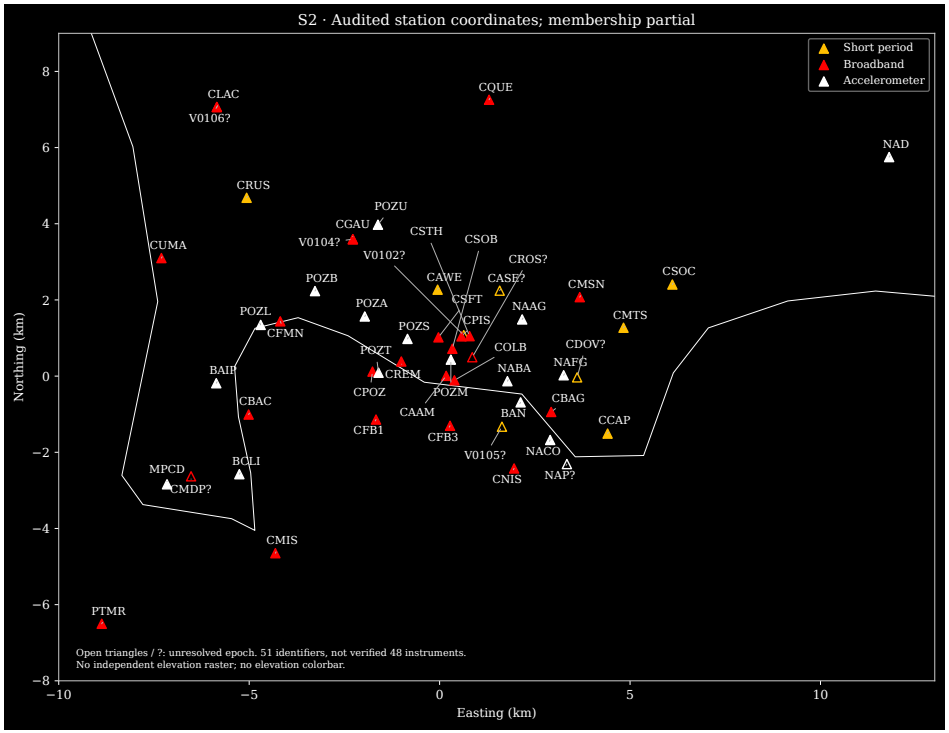
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

Independent result

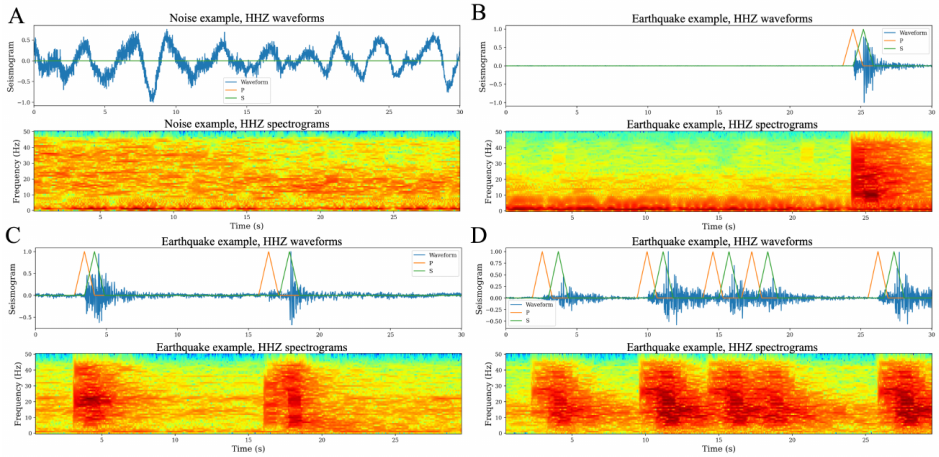
Panel S2 — partially reproduced



Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established. Colors distinguish broadband (red), short period (yellow) and accelerometer (white) instruments; no independent relief raster is available.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.

DOI: 10.1126/science.adw9038.

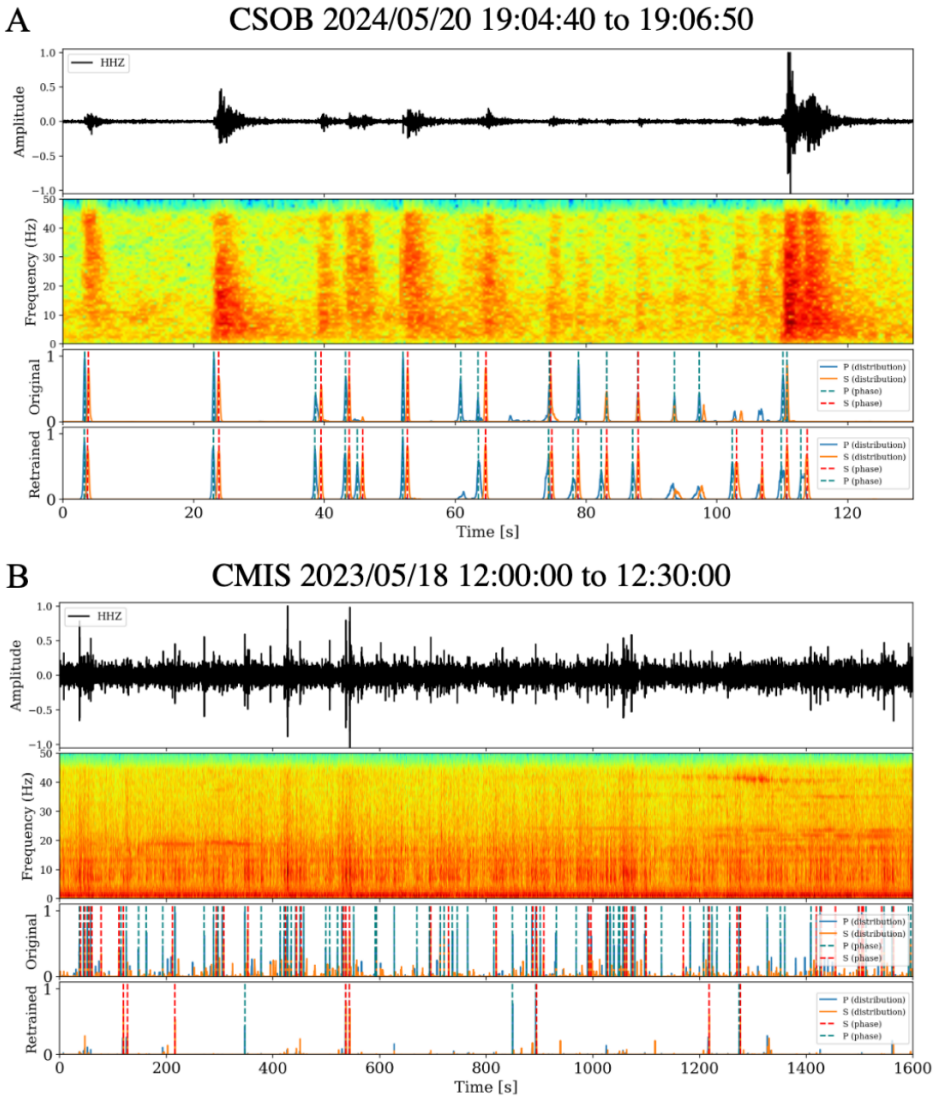
Independent result

Missing: S3A, S3B, S3C, S3D — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

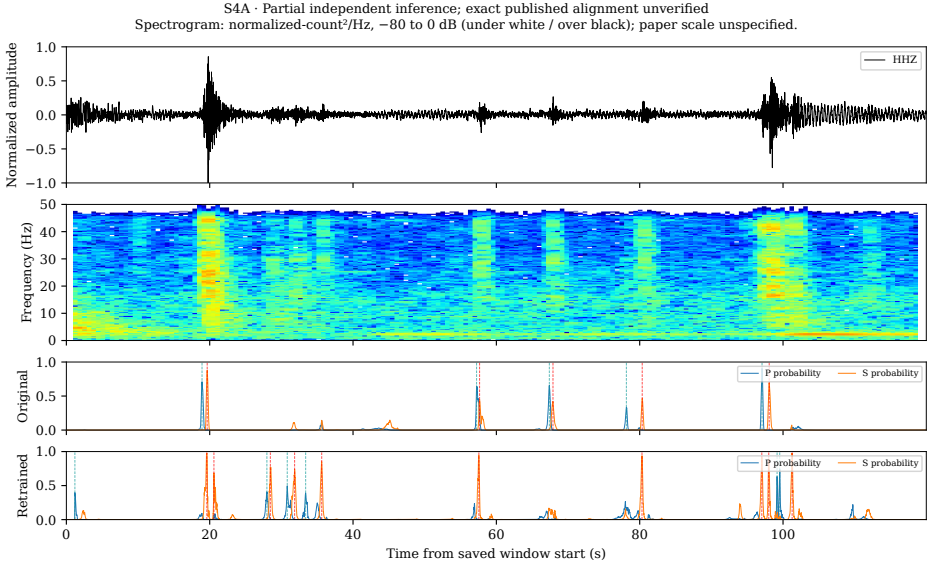
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

Independent result

Panel S4A — partially reproduced



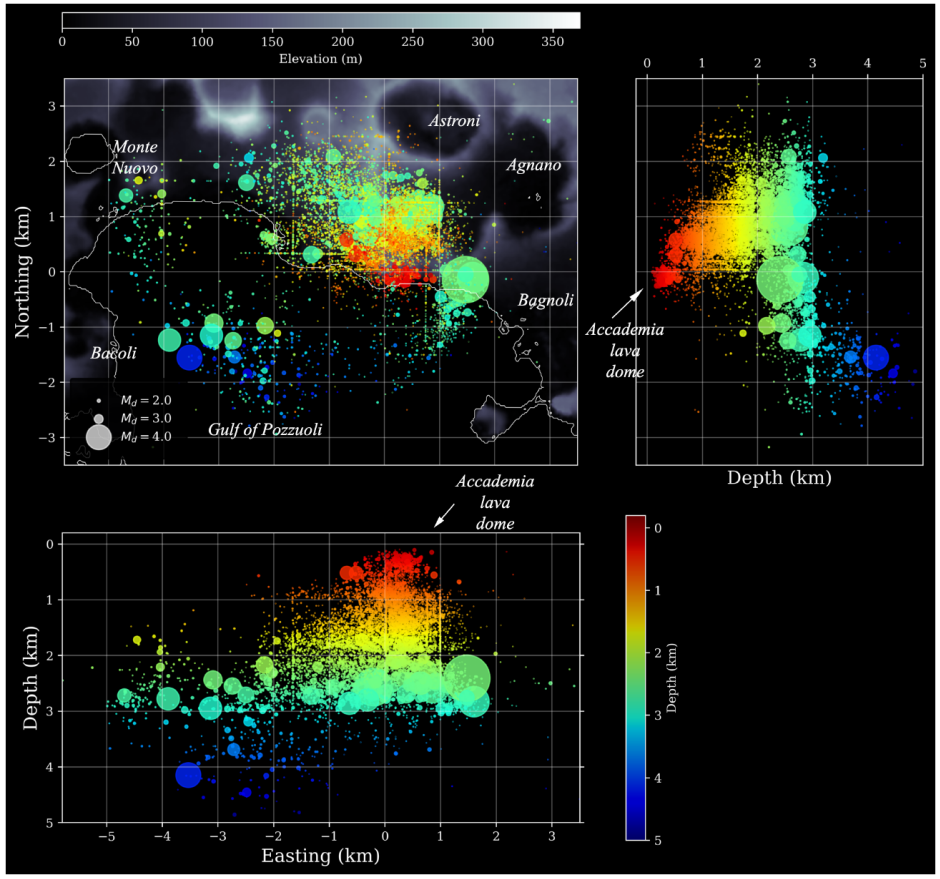
Partial S4A inference comparison on original CSOB observations, 20 May 2024 20:10:03–20:12:03 UTC. The caption's two-hour offset is interpreted literally; exact published alignment is unverified. Original weights: 5 P and 5 S; retrained weights: 6 P and 10 S. No labeled accuracy or retraining claim.

Missing: S4B — not reproduced

S4B: exact CMIS noise-window timing and reference labels are unavailable. Other calendar-window diagnostics are not inserted as that panel.

Figure S5: Routine-catalog locations

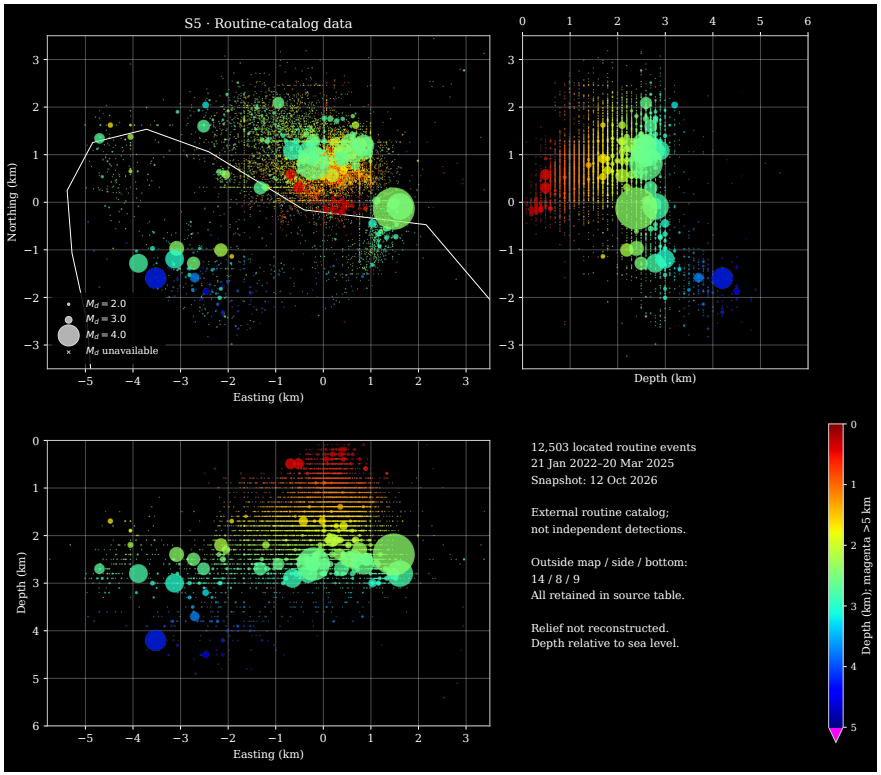
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

Independent result

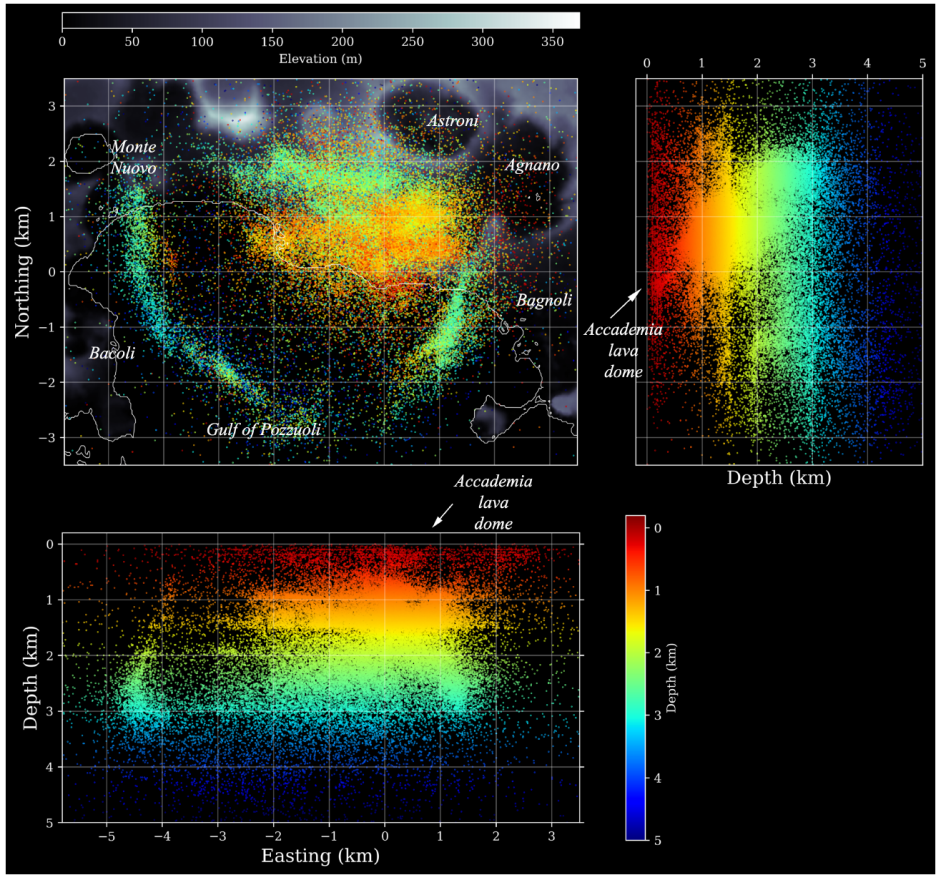
Panel S5, all views — partially reproduced



External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections. Depth colors: fixed 0–5 km, red to blue; magenta >5 km, no data clipping. S5 limits/ticks are printed; 1A endpoints are unlabeled and 0–5 km is an explicit approximation. Circle area law is an explicit display assumption, not recovered exactly. Missing magnitudes are crosses. Routine events are external context.

Figure S6: Initial VELEST locations

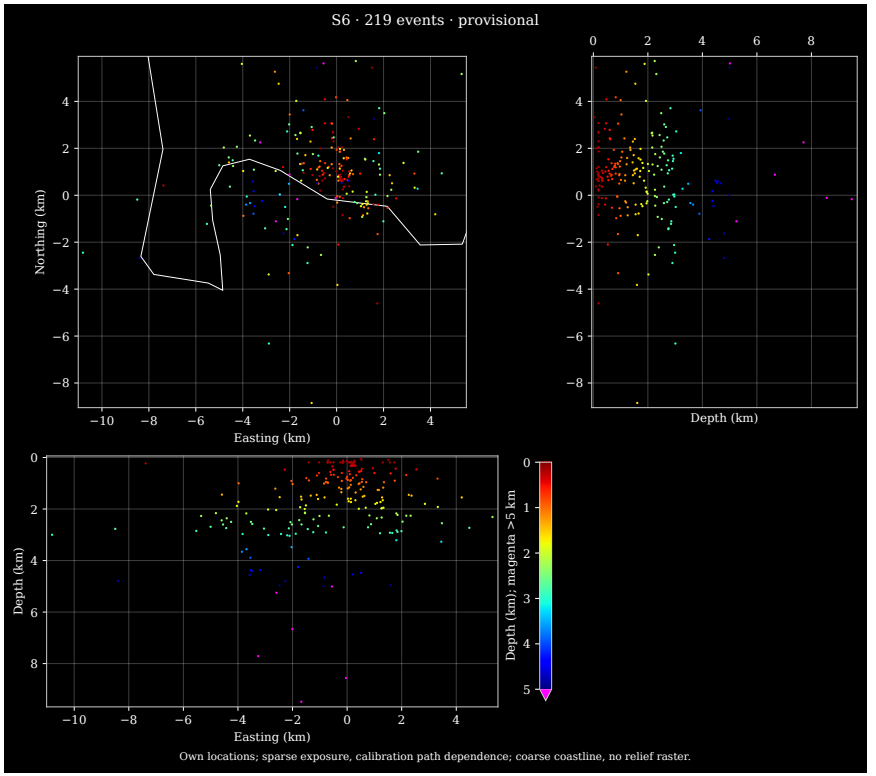
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

Independent result

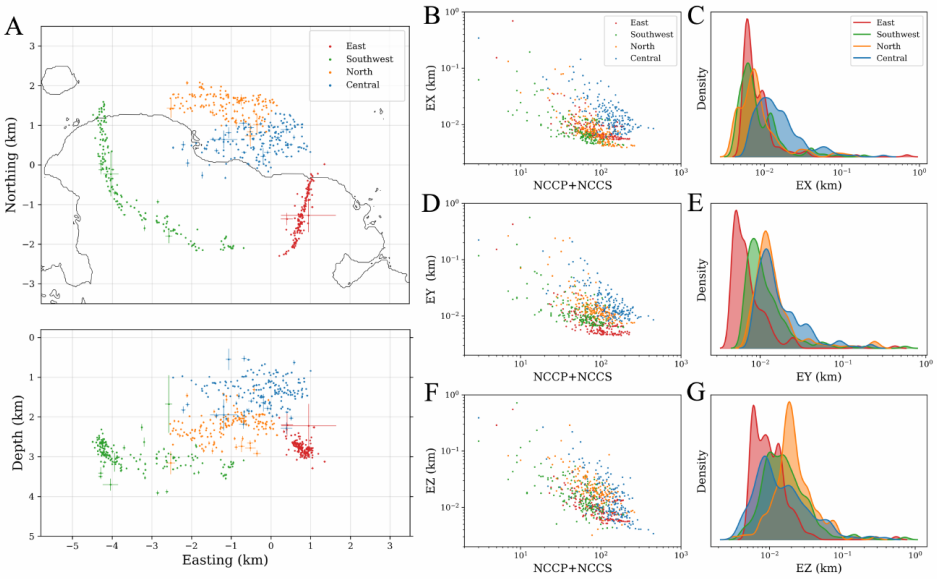
Panel S6; latest provisional numerical counterpart — partially reproduced



219 independently fitted VELEST events pass numerical pick/RMS/gap filters. Full precision paths differ by 0.89 km in depth; regional starts differ by 1.04 km. Physical ordering and small final changes do not establish a stable physical solution.

Figure S7: Location uncertainties

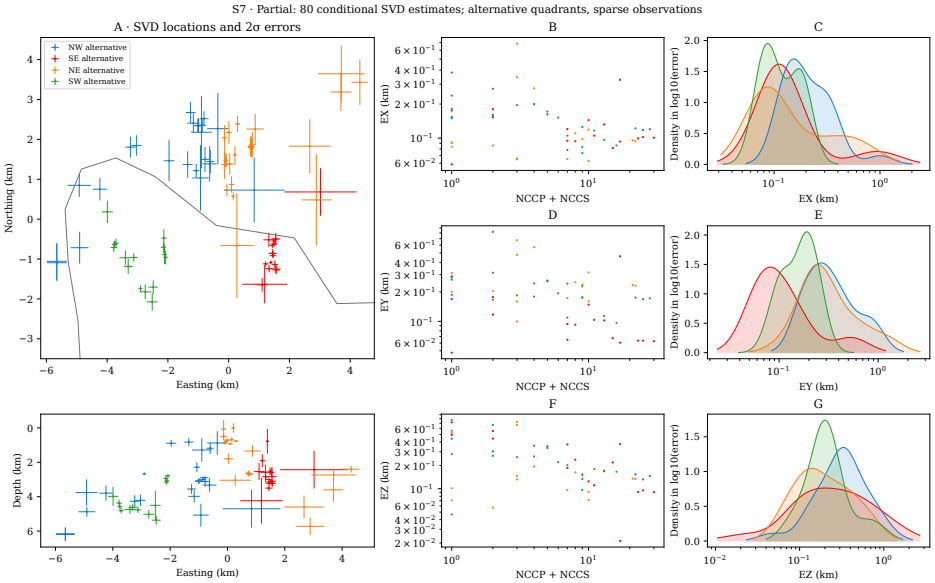
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

Independent result

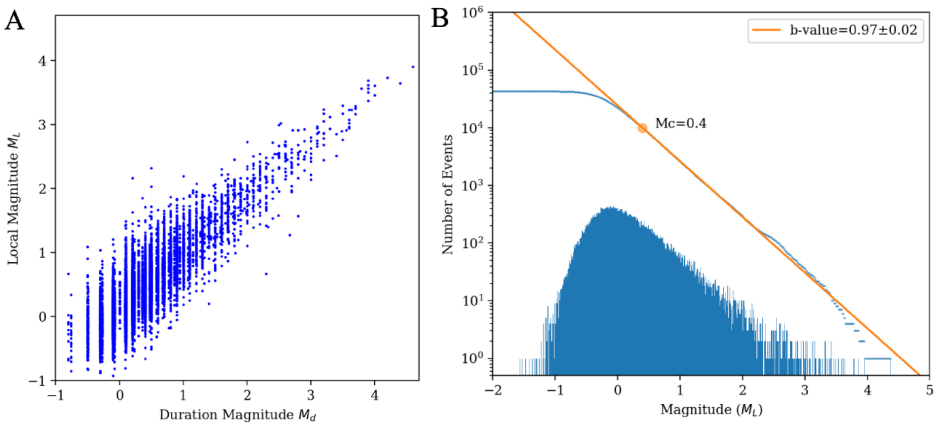
Panel S7; latest provisional numerical counterpart — partially reproduced



80 conditional HypoDD SVD estimates from every supported connected component in four alternative geographic quadrants. The largest reported vertical 2-sigma error is 1.337 km. These errors exclude model, phase-identification and coverage bias. Zero correlation counts are omitted only from logarithmic count axes.

Figure S8: Magnitude calibration and frequency distribution

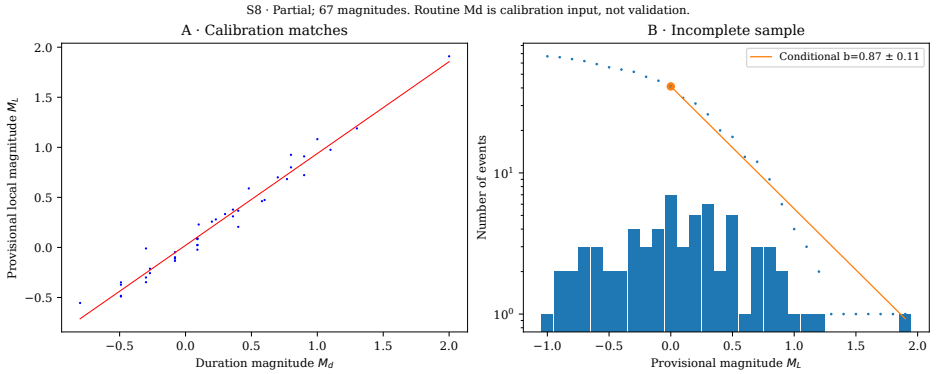
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

Independent result

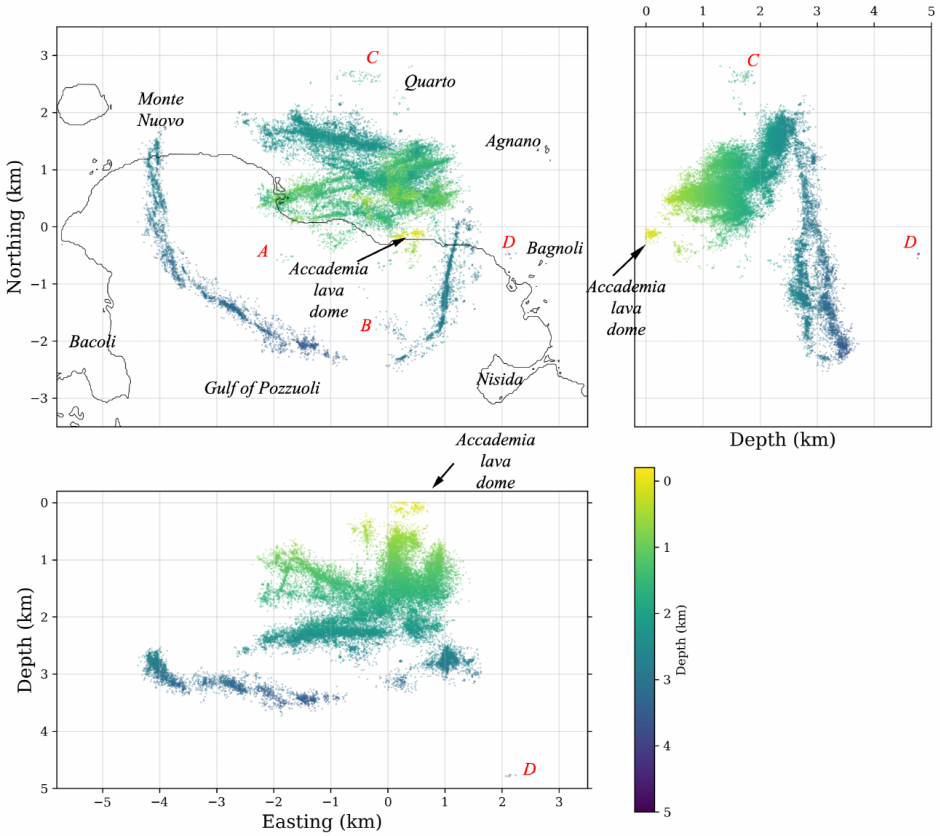
Panel S8; latest provisional numerical counterpart — partially reproduced



67 own provisional magnitudes; 41 duration-magnitude matches in A. 5887 channel measurements, 1623 above 5 dB, 884 amplitude equations. Conditional Mc=0.0 and b=0.866 +/- 0.110, N=41. Routine Md supplies calibration, not validation; discontinuous acquisition prevents population-completeness inference.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

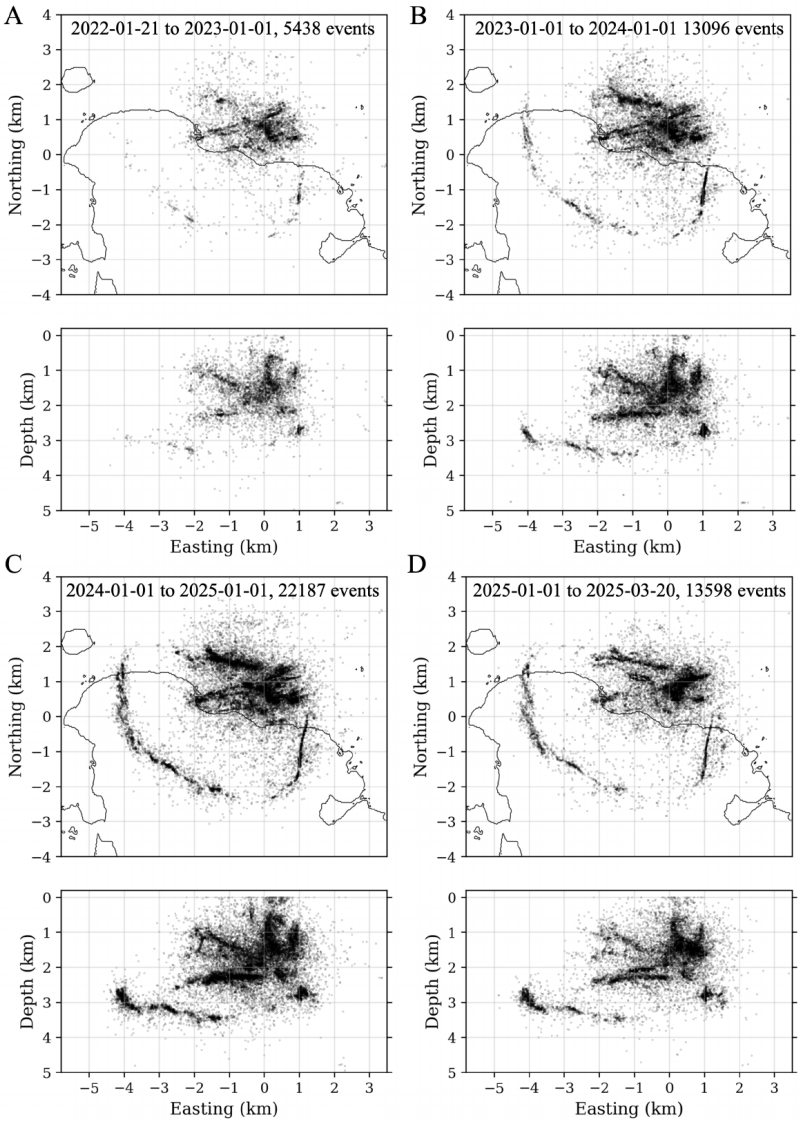
Independent result

Missing: S9/map, S9/northing-depth, S9/easting-depth — not reproduced

The documented NCCP+NCCS>100 selection was computed on the provisional relocated catalog and returned zero events. No qualified spatial result exists to plot. More waveform-linked neighboring events are needed; the threshold was not relaxed.

Figure S10: Seismicity by year

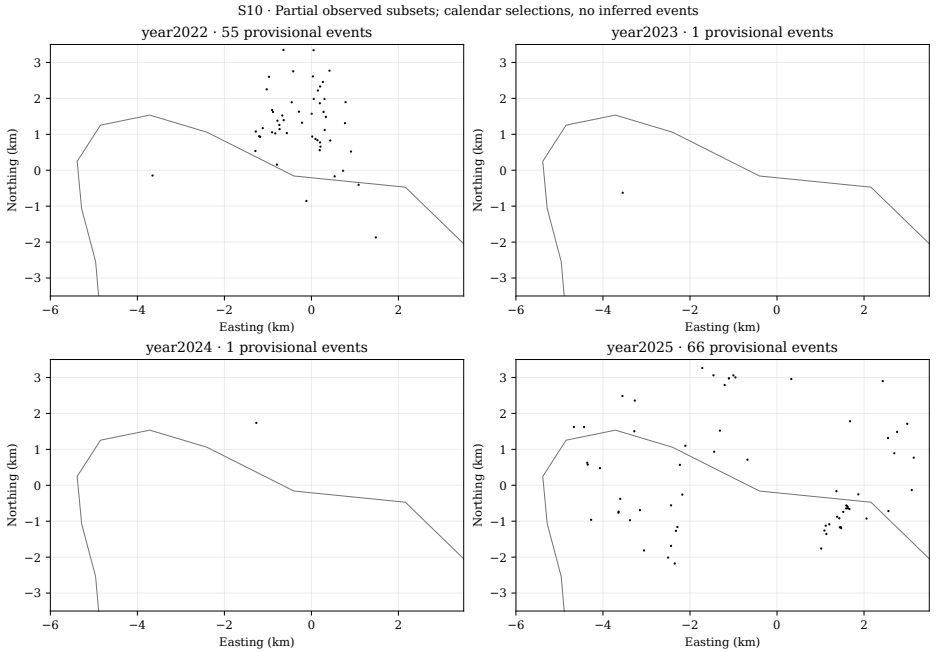
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

Independent result

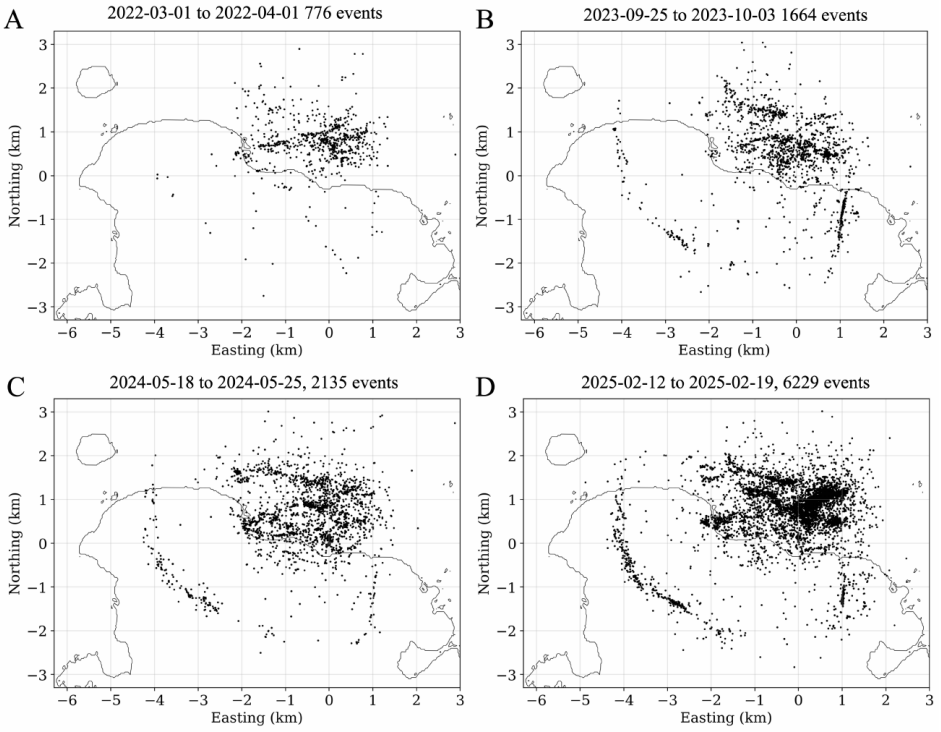
Panel S10; latest provisional numerical counterpart — partially reproduced



Own relocated calendar-year subsets: 55 in 2022, one each in 2023 and 2024, 66 in 2025. These counts reflect sparse acquisition and filtering, not annual seismicity rates.

Figure S11: Four major swarms

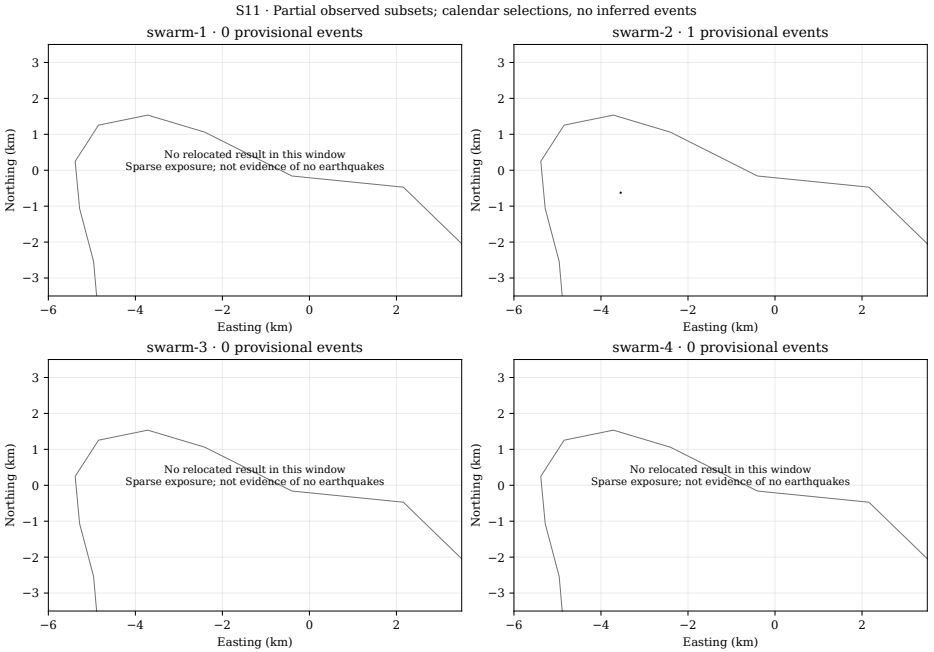
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

Independent result

Panel S11; latest provisional numerical counterpart — partially reproduced



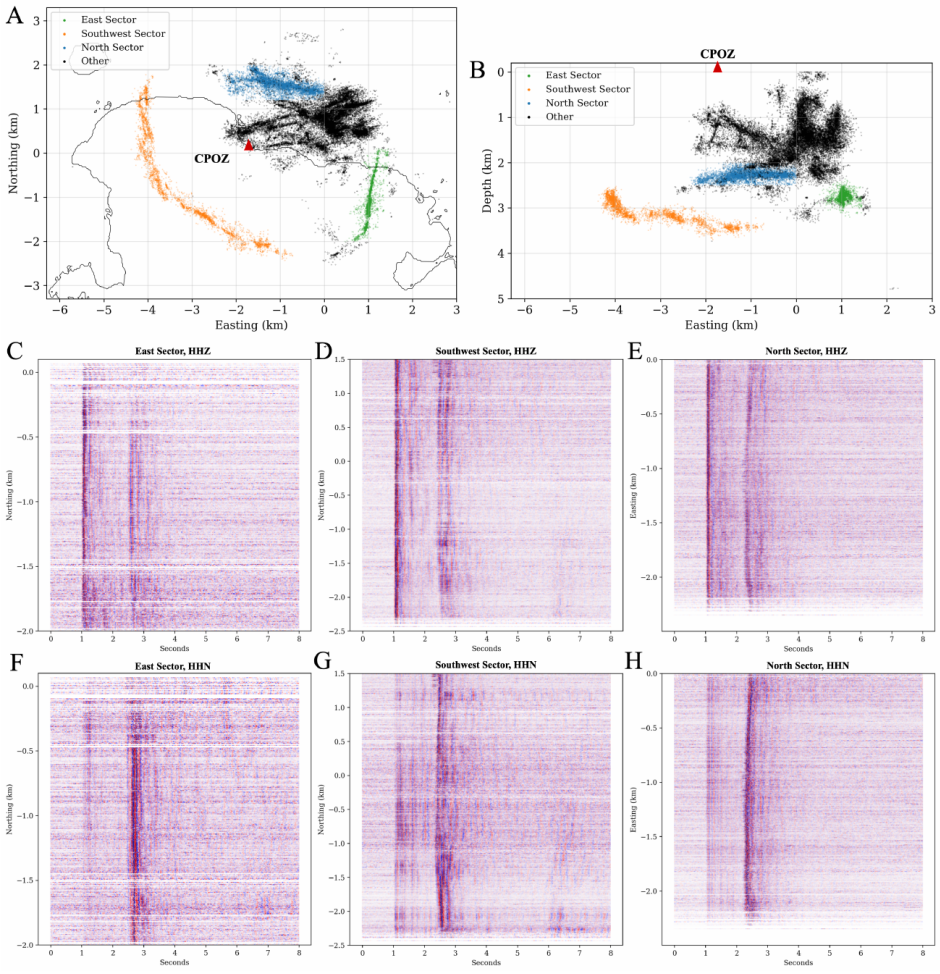
Written calendar windows, inclusive end days. Only swarm 2 contains one retained independent event; the other windows have no qualifying result. Empty selections are not evidence that earthquakes did not occur.

Missing: S11/swarm-1, S11/swarm-3, S11/swarm-4 — not reproduced

S11/swarm-1, swarm-3 and swarm-4 have no surviving independently relocated events in their written calendar windows; sparse exposure, not evidence that no earthquakes occurred.

Figure S12: Ring sectors and CPOZ common-receiver gathers

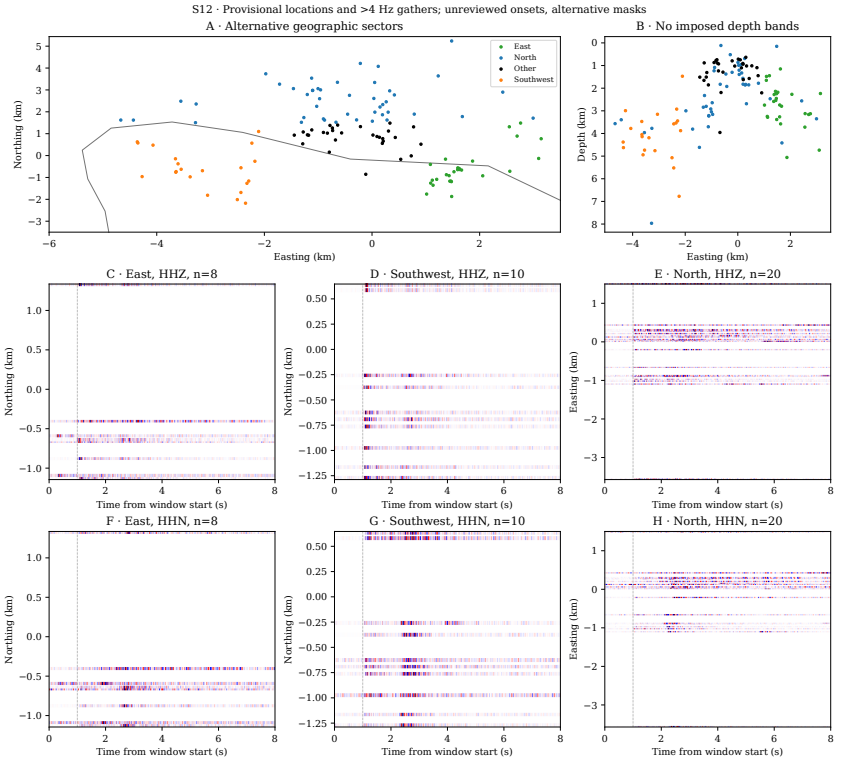
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

Independent result

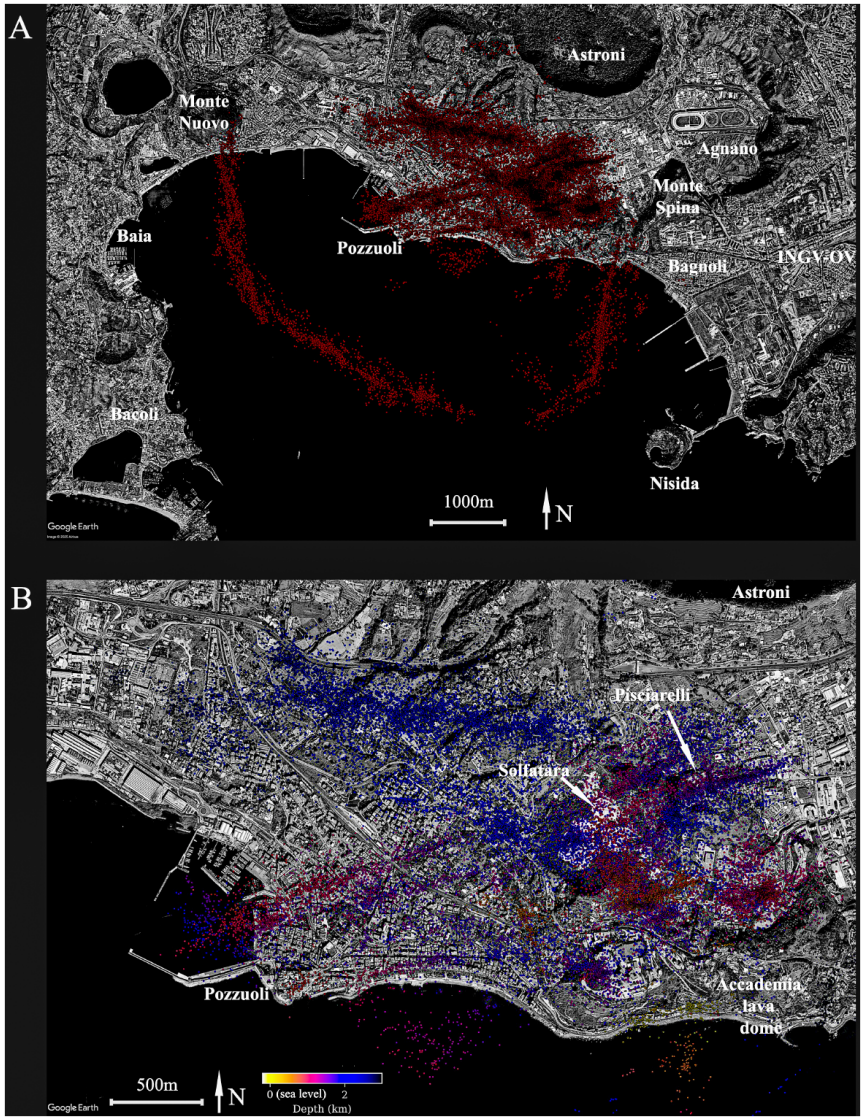
Panel S12; latest provisional numerical counterpart — partially reproduced



122 original CPOZ component traces from 61 events, aligned on own P picks and filtered above 4 Hz. Geographic sector masks are explicit alternatives. Individual amplitudes are normalized; missing traces are not interpolated. Manual onset/depth validation remains incomplete.

Figure S13: Relocated seismicity on satellite imagery

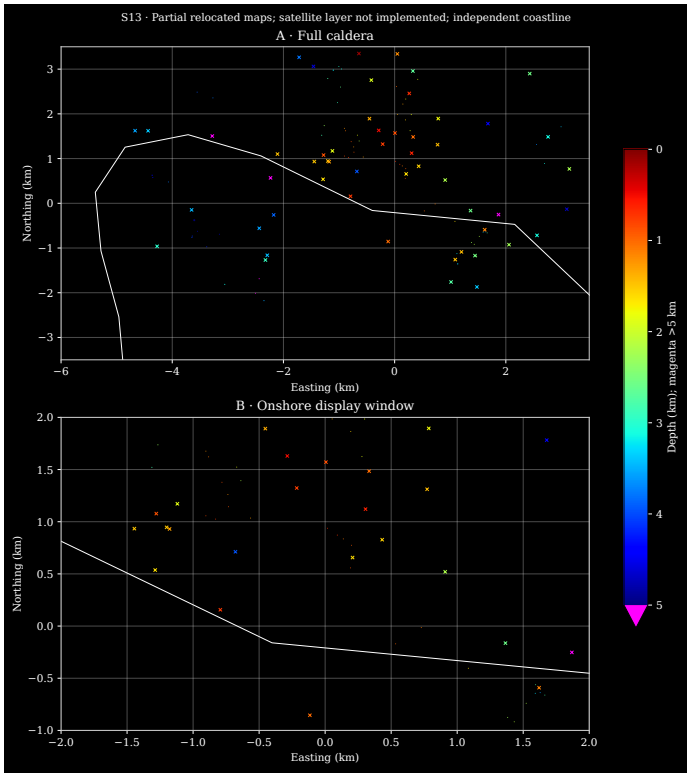
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

Independent result

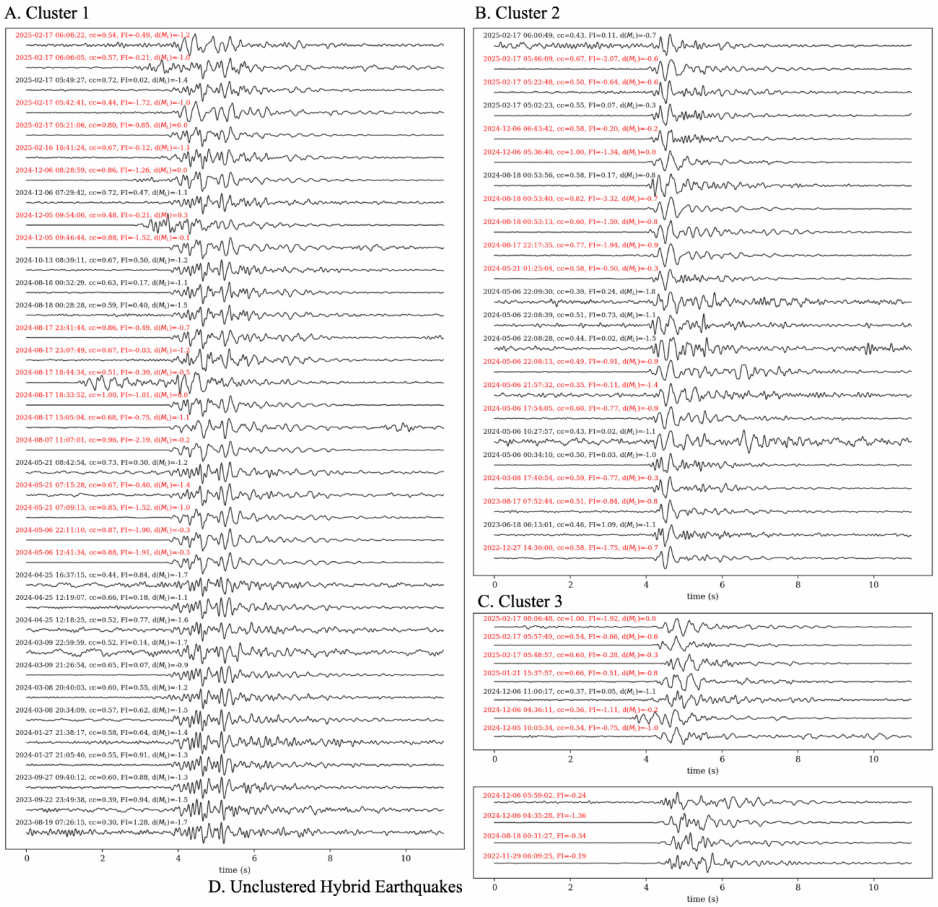
Panel S13; latest provisional numerical counterpart — partially reproduced



Latest own provisional relocated coordinates in whole-caldera and central views. An independent coarse coastline replaces the unimplemented satellite layer. No published pixels enter the maps.

Figure S14: Hybrid-event clusters

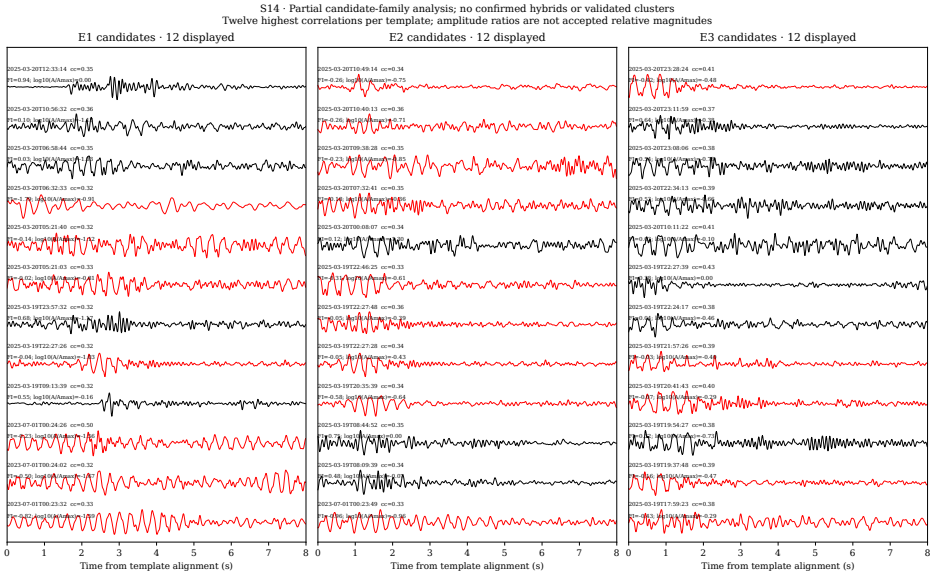
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

Independent result

Panel Template-family candidates; not confirmed clusters — partially reproduced



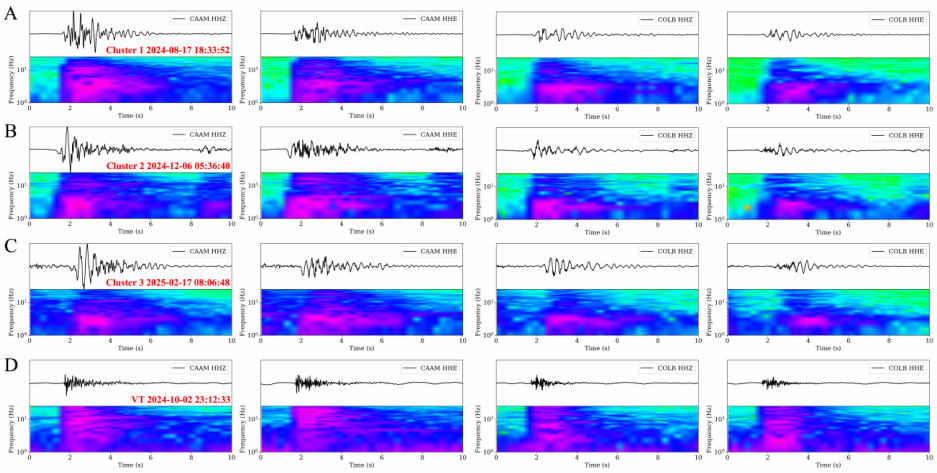
Measured waveforms, frequency indices and amplitude ratios for the twelve highest correlations per template. These candidates are not confirmed hybrids or validated clusters.

Missing: S14/unclustered — not reproduced

S14/unclustered: no independent unclustered hybrid sample or validated family assignment. The three-template threshold scan cannot itself supply that class.

Figure S15: Hybrid and shallow-event waveforms

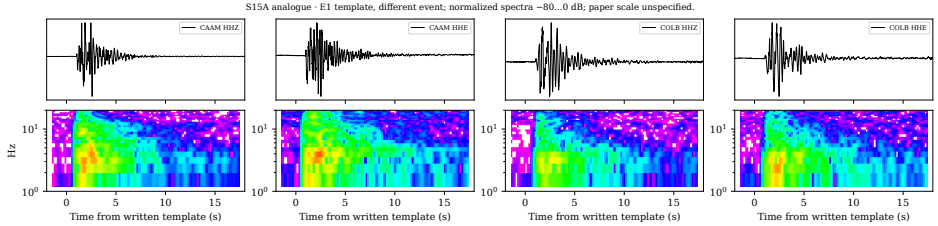
Published reference



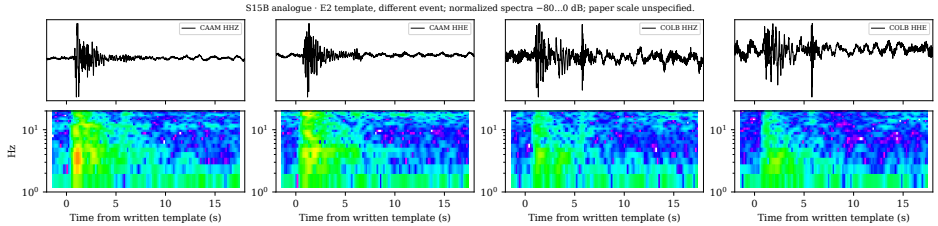
Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

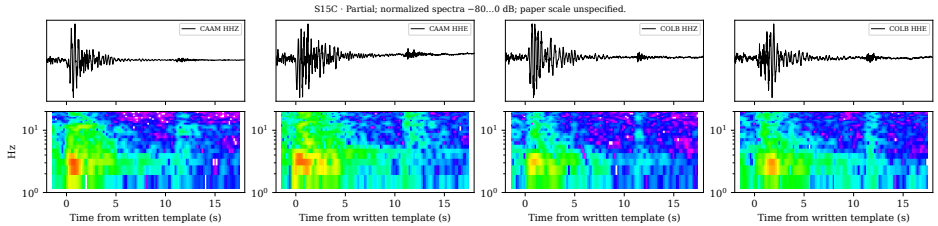
Panel S15A analogue: different documented E1 event — partially reproduced



Panel S15B analogue: different documented E2 event — partially reproduced



Panel S15C: documented E3 event — partially reproduced



A/B use explicitly different E1/E2 examples; C uses the documented E3 template. All traces and spectra were computed from original observations. No qualified D example is available.

Missing: S15D — not reproduced

S15D: no retained independently relocated event with depth <0.6 km south of 40.83 N in this cohort, hence no suitable west-Accademia extremely-shallow example. Incorrectly labeled examples are not substituted.